National University
Of Computer and Emerging Sciences

Foundation of Advancement
Of Science and Technology

Lab Number:	05
Course Code:	CL2001
Course Title:	Data Structures - Lab
Semester:	Fall 2026
Instructor:	Muhammad Mehdi
Email:	muhammad.mehdi@nu.edu.pk

Guidelines

- You may use **only the concepts covered in the lab** to complete this task.
- Your submission must be a **single PDF file** that includes:
  - The source code (in plain text)
  - Screenshots of the corresponding outputs
- Name your file using the following format: **rollno_name_labtasknumber.pdf**
  Example: *25p-1234_ozair_labtask05.pdf*
- Unethical use of AI tools will result in **0 marks**.
- Late submissions will incur a penalty of **-3 marks per day**.

Tasks

ONLY USE POINTER NOTATION TO IMPLEMENT ALL THE SOLUTIONS. ARRAY INDEXING [ ] IS NOT ALLOWED.

1 Singly Linked List to Doubly Linked List

Modify the given implementation by defining an additional constructor that accepts a **Singly Linked List** object as its parameter and creates a corresponding doubly linked list from it.

The constructor should traverse the given singly linked list and create a new doubly linked list containing the **same elements in the same order**. The `prev` and `next` links of each node must be correctly established.

Example:

Given singly linked list: `9->8->7->6->End`

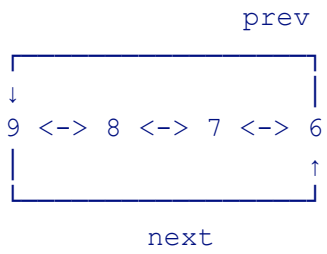
The resulting doubly linked list should be: `End<-9<->8<->7<->6->End`

2 Implement a Circular Doubly Linked List

Implement a **Circular Doubly Linked List** by modifying or extending the provided **Doubly Linked List** or the provided **Circular Singly Linked List** implementation.

The circular doubly linked list should maintain both `next` and `prev` links such that the list forms a circular structure.

For example, for the elements `9, 8, 7, 6`:



In other words:

- The `next` pointer of the last node should point to the first node.
- The `prev` pointer of the first node should point to the last node.
- All other `next` and `prev` pointers should correctly connect adjacent nodes.
- The list should remain circular after insertion and deletion operations.

Implement the required functionality while ensuring that the circular structure is correctly maintained after every operation.